4.2 Data Analysis

4.3 Simulation of plate bending

Simulating plate bending is "cheaper" than making experiments, in time and resources. The simulation is used for predicting the results from the experiments. The simulation software used is LS-Dyna and the following chapter will explain in detail the keyword file and how multiple simulations are run.

4.3.1 Finite theory

4.3.2 LS-Dyna keyword

In LS-Dyna a simulation is controlled via keywords in a keyword file "main.k", the content of the file controls every aspect of how the simulation is run. A model part, a physics part and a solver part, this is shown in fig. 4.4. All input files (*.k) required to reproduce the simulations are available in the project repository: `https://github.com/node997/dyna_pladebuk` (Project Group 3.220 2026).

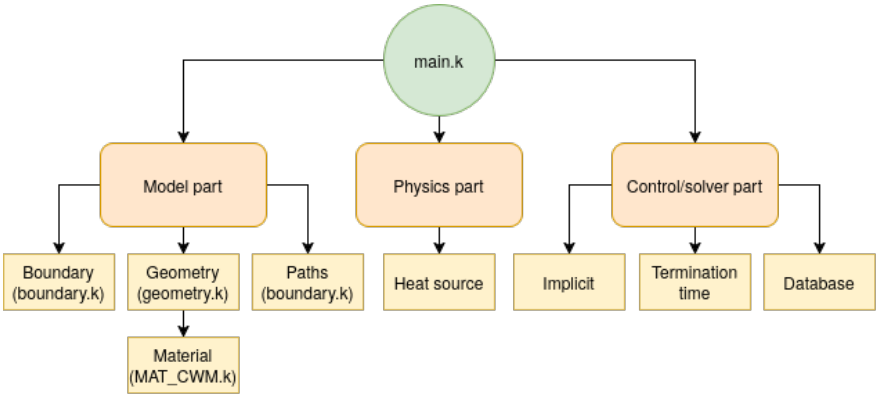

Figure 4.4: Ls-dyna build of 3 parts, a model part, a physics part and a control/solution part which is seperated into files for a more manageble "main.k" file, this shows how the "main.k" file is the binding joint for every file.

This is however just broadly how the simulation is being controlled, since the physics and solver parts are build with keywords and needs to be configured in order to work for all the following, a zero means standard LS-Dyna.

Model keywords

FOr defining the boundary condition of the plate being fixed at one end the following keyword is used. NSID is the id or name for the fixture, this is used when defining how this effects the simulation. CID is the coordinate system id which is called 0. DOFX, DOFY,

DOFZ, DOFRX, DOFRY and DOFRZ set to 1 means that it cannot move or rotate in any of these axes.

```
*BOUNDARY_SPC_SET
$#     nsid       cid      dofx      dofy      dofz     dofrx     dofry     dofrz
         2         0         1         1         1         1         1         1
```

For defining the boundary node set the following keyword is used. SID is for defining what node set that is in use for the boundary fixed list SID is set to 2. DA1 is for the first nodal attribute to the system this goes all the way up to the fourth node at DA4 these are set to 0. SOLVER is for the solver type which is set to MECH for mechanical. ITS is for specifying what the type for the coupling is used, this is set to 1 for tied contact coupling. This node set is called "NODESET (SPC) 2".

```
*SET_NODE_LIST_TITLE
NODESET(SPC) 2
$#     sid       da1       da2       da3       da4    solver       its         -
         2       0.0       0.0       0.0       0.0MECH               1
```

For setting the heat source path a list of nodes is neccesary, this is defined in the same way as the keyword above but with a different SID.

```
*SET_NODE_LIST_TITLE
weld_path_nodes
$#     sid       da1       da2       da3       da4    solver       its         -
        20       0.0       0.0       0.0       0.0MECH               1
$#    nid1      nid2      nid3      nid4      nid5      nid6      nid7      nid8
```

This keyword is the same as the ones above except this is for the segments surrounding the node path but with a different SID.

```
*SET_SEGMENT_TITLE
weld_path_segments
$#     sid       da1       da2       da3       da4    solver       its         -
       569       0.0       0.0       0.0       0.0MECH               0
$#      n1        n2        n3        n4        a1        a2        a3        a4
```

This keyword is the same is the one above with a different SID.

```
*SET_SEGMENT_TITLE
fconv
$#     sid       da1       da2       da3       da4    solver       its         -
    420420       0.0       0.0       0.0       0.0MECH               0
$#      n1        n2        n3        n4        a1        a2        a3        a4
```

For including material properties a MAT_CWM.k file is included with the following attributes.

```
*INCLUDE
MAT_CWM.k
```

MAT_CWM(MAT_270), where MAT is of type 270 which is a thermo-elastic-plastic model that allows for annealing triggered by temperature. CWM stands for computational welding mechanics. MID is the material ID this is set to 1 and is an arbitrary number. RO is

the density of the material. LCEM is the load curve for Youngs modulus as a function of temperature. LCPR is the load curve for Poissons ratio as a function of temperature. LCSY is load curve for yield stress as a function of temperature. LCHR is the load curve for the hardening modulus as a fuctions of temperature. LCAT is load curve for thermal expansion coefficient as function of temperature. BETA is 0 for kinematic hardening. TASTART is annealing temperature start. TAEND is annealing temperature end. TLSTART is birth temperature start. TLEND is the birth temperature end. EGHOST is ghost materials youngs modulus. PGHOST is Poissons ratio for the ghost material. T2PHASE is the temperature when there is a phase change. T1PHASE is when this phase change ends. ANOP is anealling according to thermal expansion, set to 0 means there is no modification with anealing. POSTV is a post processing flag that can be tweaked for adding history to the material.

```
*MAT_CWM_TITLE
Plates_Mech
$#     mid        ro      lcem      lcpr      lcsy      lchr      lcat      beta
         17.90000E-9       411       412       413       414       415       0.0
$# tastart     taend   tlstart     tlend    eghost    pghost    aghost
    1331.0    1773.0-1.00000E8  -1000000   10000.0     0.2841.0000E-20
$# t2phase   t1phase     anopt     postv
       0.0       0.0       0.0         0
```

MAT_THERMAL_CWM(MAT_T07) is the thermal material with temperature dependencies. TMID is the id for the material. TRO is the thermal density, this is set to 0 for the default structural density. TGRLC is the thermal generation rate curve being a constant multiplier value when set to 0. TGMULT is the thermal generation multiplier and this is set to 0 for no heat generation. HDEAD is specific heat for the material when its inactive before birth. TDEAD is thermal conductivity for the inactive material before birth. TLAT is the temperature for phase change. HLAT is the latent heat. LCHC is the load curve for specific heat as a function of temperature. LCTC is the load curve for thermal conductivity as a function of temperature. TLSTART is the burth of material. TLEND is the birth of temperature material end. TISTART is birth time start. TIEND is birth time end. HGHOST is specific heat of the ghost material, and TGHOST is the thermal conductivity for ghost material.

```
*MAT_THERMAL_CWM_TITLE
Plates_thermal
$#    tmid       tro     tgrlc    tgmult     hdead     tdead      tlat      hlat
         1       0.0       0.0       0.0       0.0       0.0       0.0       0.0
$#    lchc      lctc   tlstart     tlend   tistart     tiend    hghost    tghost
       211       212   -1000.0   -1000.0       0.0       0.06.500000E8     0.001
```

The following curves are what the SIDs are refering to in the keywords above. With temperature being on the right as a1 and o1 being the result

```
*DEFINE_CURVE_TITLE
LCHC - specific heat as function of temperature
$#    lcid      sidr       sfa       sfo      offa      offo    dattyp     lcint
       211         0       1.0       1.0       0.0       0.0         0         0
```

```
$#                a1                  o1
                 0.0    4.6200000000e+08
               273.0    4.6200000000e+08
               373.0    4.9600000000e+08
               473.0    5.1200000000e+08
               573.0    5.2500000000e+08
               673.0    5.4000000000e+08
               873.0    5.7700000000e+08
              1073.0    6.0400000000e+08
              1473.0    6.7600000000e+08
              1773.0    7.0000000000e+08
              9773.0    7.0000000000e+08
```

```
*DEFINE_CURVE_TITLE
LCTC - thermal conductivity
$#    lcid      sidr       sfa       sfo      offa      offo    dattyp     lcint
       212         0       1.0       1.0       0.0       0.0         0       200
$#                a1                  o1
                 0.0                14.6
               273.0                14.6
               373.0                15.1
               473.0                16.1
               573.0                17.9
               673.0                18.0
               873.0                20.8
              1073.0                23.9
              1473.0                32.2
              1773.0               120.0
              9773.0               120.0
```

```
*DEFINE_CURVE_TITLE
LCEM - Youngs modulus as temperature
$#    lcid      sidr       sfa       sfo      offa      offo    dattyp     lcint
       411         0       1.0       1.0       0.0       0.0         0         0
$#                a1                  o1
                 0.0            198500.0
               273.0            198500.0
               373.0            193000.0
               573.0            176000.0
               673.0            167000.0
               873.0            159000.0
              1073.0            151000.0
              1473.0             60000.0
              1773.0             10000.0
              9773.0             10000.0
```

```
*DEFINE_CURVE_TITLE
LCPR - Poissons ratio as function of temperature
$#    lcid      sidr       sfa       sfo      offa      offo    dattyp     lcint
       412         0       1.0       1.0       0.0       0.0         0         0
$#                a1                  o1
                 0.0               0.294
               273.0               0.294
               373.0               0.295
               573.0                0.31
               673.0               0.318
               873.0               0.326
```

```
                1073.0                0.333
                1473.0                0.339
                1773.0                0.388
                9773.0                0.388
```

```
*DEFINE_CURVE_TITLE
LCSY - yield stress to temperature
$#     lcid      sidr       sfa       sfo      offa      offo    dattyp     lcint
       413          0       1.0       1.0       0.0       0.0         0         0
$#                 a1                  o1
                  0.0             265.0
                273.0             265.0
                373.0             218.0
                573.0             170.0
                673.0             155.0
                873.0             149.0
               1073.0              91.0
               1473.0              25.0
               1773.0              10.0
               9773.0              10.0
```

```
*DEFINE_CURVE_TITLE
LCHR - hardening modulus
$#     lcid      sidr       sfa       sfo      offa      offo    dattyp     lcint
       414          0       1.0       1.0       0.0       0.0         0         0
$#                 a1                  o1
                  0.0             350.0
                273.0             350.0
                373.0             300.0
                473.0             250.0
                573.0             150.0
                673.0             100.0
                873.0              55.0
               1073.0              10.0
               1773.0               1.0
               9773.0               1.0
```

```
*DEFINE_CURVE_TITLE
LCAT - thermal expansion
$#     lcid      sidr       sfa       sfo      offa      offo    dattyp     lcint
       415          0       1.0       1.0       0.0       0.0         0         0
$#                 a1                  o1
                  0.0    1.7000000298e-05
                273.0    1.7000000298e-05
                373.0    1.7399999706e-05
                573.0    1.8599999748e-05
                673.0    1.9099999918e-05
                873.0    1.9600000087e-05
               1073.0    2.0199999199e-05
               1473.0    2.0699999368e-05
               1773.0    2.1600000764e-05
               9773.0    2.1600000764e-05
```

For defining the plate the following keyword is used. PID is the part id this is set to 1. SECID is the section identification which is from the "*SECTION_SOLID" keyword these need to match to make sure that the correct parts have the designated material. MID is

the material ID for using the steel material that is written in MAT_CMW.k MID needs to match. EOSID is for using preasure this is not used in this simulation therefore this is 0. HGID is hourglass viscosity and is defined (if used) in *HOURGLASS since it is not in use this is set to 0. ADPOPT is for indicating if this part is adapted. This is not used so adaptive remeshing is turned off indicated by it being 0. TMID is the thermal material id and needs to match from the MAT_THERMAL_CWM(MAT_T07) therefore this is set to 1.

```
*PART
$#                                                                      title
Plade
$#     pid     secid       mid     eosid      hgid      grav    adpopt      tmid
         1         1         1         0         0         0         0         1
```

The following keyword is for defining what type of element formulation is used. For solid elemets _SOLID is used. SECID is the ID for the section and is used ind *PART. ELFORM is the type of elemetn formulation. By setting ELFORM to 2 8 point hexahedron are used. AET is 0 for non ambient element type. COHOFF is the default which means the cohisive layer is between the top and the bottom. GASKEIT is only used if *MAT_COHESIVE _GASKET is defined, therefore this is 0.

```
*SECTION_SOLID
$#   secid    elform       aet    unused    unused    unused    cohoff   gaskeit
         1         2         0                                     0.0       0.0
```

For including the mesh a geometry.k file is made seperate from the main.k to keep everything organized.

```
$# 2mm kvadratiske elementer
*INCLUDE
geometry.k
```

Physics keywords

For cooling the plate a convection is made for blowing air on the plate after cooling. SSID is the id for the convection. PSEROD is for updating segments exposed to the environment, this is turned off by PSEROD being set to 0. HLCID is the load curve for controlling when to turning on or off the convection. HMULT is the convection heat transfer coefficeint, h. TLCID is envoriment temperature. By setting this to 0 the temperature is defined by a constant TMULT. TMULT is the enviroment temperature , multiplier. By setting this ti 293K the ambient temperature of the model is 22 C°. LOC is for shells and is not used.

```
*BOUNDARY_CONVECTION_SET
$#    ssid    pserod
    420420         0
$#   hlcid     hmult     tlcid     tmult       loc
      6969      0.02         0     295.0         0
```

For emulating a moving heat source the following keyword is used. This card is highly custimizable from heat source type to its power, and efficiency. It is used for welding but

can be used for this application as well. The moving heat source will be called "welding" for following sections. PID is the part ID for where the welding happens. PTYP os for what type of PID, with this being set to 1 its for a part ID. NSID1 is the weld path this is a defined node set with ID 20. SPD1 is the speed of the weld in mm/s. NSID2 is the segment set/ID containing the nodes from NSID1 this is segment id 569. SPD2 is speed reference and this is ignored since it is 0. NCYC is the number of sub steps, this allows for thermal dumping. RELEVEL redefines SPD1 and SPD2 to be absolute speeds. IFORM is the type of energy rate density distrobution with 1 being goldak-type heat source. LCID is the load curve id for weld energy input rate as function of time. Q is the curve multiplier. LCROT is the angle of how the weld is aimed towards the plate. LCLAT is for lateral offset as a function of time, this is not used. DISC is resolution for accurate integration. ENFOR is for heat input enforcement, and is not used. P_i is for how the IFORM is formed (in mm)

```
*BOUNDARY_THERMAL_WELD_TRAJECTORY
$#     pid      ptyp     nsid1      spd1     nsid2      spd2      ncyc    relvel
         1         1        20       3.0      -569       0.0        10         0
$#   iform      lcid         q     lcrot     lcmov     lclat      disc     enfor
         1      1010       1.2      1011      1012         0       2.0         0
$#      p1        p2        p3        p4        p5        p6        p7        p8
      16.0       3.0      16.0      16.0       0.6       1.4       3.0       0.0
```

For having gravity acting on the plate this is defined in the following keyword. LCID is the id for this. SF is the load curve scale factor. CID is the global coordinate system for the simulation.

```
*LOAD_BODY_Y
$#    lcid        sf    lciddr        xc        yc        zc       cid
         3       1.0         0       0.0       0.0       0.0        23
```

For setting the initial temperature for the system the following keyword is defined.

```
*INITIAL_TEMPERATURE_SET
$#    nsid      temp       loc
         0     295.0         0
```

For defining the global coordinate system the following keyword is used. CID is the coordinate id. CL is x coordinate of a point in the local x-axis, YP is a point in the xy plane.

```
*DEFINE_COORDINATE_SYSTEM
$#     cid        xo        yo        zo        xl        yl        zl      cidl
        23       0.0       0.0       0.0       1.0       0.0       0.0         0
$#      xp        yp        zp                                             unused2
       0.0       1.0       0.0
```

Gravity load curve.

```
*DEFINE_CURVE
$#    lcid      sidr       sfa       sfo      offa      offo    dattyp     lcint
         3         0       0.0       0.0       0.0       0.0         0         0
$#                a1                  o1
                 0.0               9806.65
                 1.0               9806.65
```

```
                30.0              9806.65
               300.0              9806.65
```

The following curves are for defining how loading occurs in the welding segment of the simulation.

```
*DEFINE_CURVE
$#     lcid      sidr       sfa       sfo      offa      offo    dattyp     lcint
        21         0       1.0       1.0       0.0       0.0         0         0
$#                a1                  o1
                 0.0                 0.0
                 0.1                 1.0
                20.0                 1.0
                20.1                 0.0
              1000.0                 0.0
```

```
*DEFINE_CURVE_TITLE
LCID - THERMAL WELD TRAJECTORY
$#     lcid      sidr       sfa       sfo      offa      offo    dattyp     lcint
      1010         0       1.0       1.0       0.0       0.0         0         0
$#                a1                  o1
                 0.0                 0.0
                 0.2             1000000
                33.3             1000000
                33.4                 0.0
              1000.0                 0.0
```

```
*DEFINE_CURVE_TITLE
LROT - THERMAL WELD TRAJECTORY verification is needed !!!!
$#     lcid      sidr       sfa       sfo      offa      offo    dattyp     lcint
      1011         0       1.0       1.0       0.0       0.0         0         0
$#                a1                  o1
                 0.0                 0.0
              1000.0                 0.0
```

```
*DEFINE_CURVE_TITLE
LCMOV - THERMAL WELD TRAJECTORY !!!!
$#     lcid      sidr       sfa       sfo      offa      offo    dattyp     lcint
      1012         0       1.0       1.0       0.0       0.0         0         0
$#                a1                  o1
                 0.0                 0.2
              1000.0                 0.2
```

```
*DEFINE_CURVE_TITLE
DT_MAX - variabelt dt styret af svejsefase
$#     lcid      sidr       sfa       sfo      offa      offo    dattyp     lcint
      4441         0       1.0       1.0       0.0       0.0         0         0
$#                a1                  o1
                 0.0                 0.1
                50.0                 0.1
                60.1                 0.5
               120.0                 0.5
```

```
*DEFINE_CURVE_TITLE
AFKOLING - turns on cooling after xxx seconds
$#     lcid      sidr       sfa       sfo      offa      offo    dattyp     lcint
```

```
      6969         0       1.0       1.0       0.0       0.0         0         0
$#                a1                  o1
                 0.0                 0.0
                50.0                 0.0
                50.1                 1.0
              1000.0                 1.0
```

Control/solver keywords

This keyword is just for the title for the simulation

```
*KEYWORD
*TITLE
$#                                                                      title
LS-DYNA SIMULERING AF VARME DER BUKKER PLADE
```

This keyword with dflag set to "1" is for the controlling duplicate parameters definitions, 4 is for accepting this silently.

```
*PARAMETER_DUPLICATION
$#   dflag
         4
```

This keyword is for defining the accuracy of the simulation. OSU=1 enables second-order objective stress update, meaning stresses are calculated at the mid-point geometry instead of the beginning of each timestep, providing a more physically accurate model. INN=4 determines the local coordinate system for all element types, enabling invariant node numbering which ensures the local coordinate system remains consistent this prevents hourglass modes from affecting material directions and maintains stable calculations over long time periods. IACC=1 enables implicit accuracy corrections, ensuring that the larger timesteps typical of implicit analysis do not compromise result quality.

```
*CONTROL_ACCURACY
$#     osu       inn    pidosu      iacc
         1         4         0         1
```

This keyword is for controlling shock waves, since there are no extreme shock values in the simulation these are all set to the defualt.

```
*CONTROL_BULK_VISCOSITY
$#      q1        q2      type
       1.5      0.06         1
```

For controlling how computation is calculated for contact surfaces. For the first part(Card 1) all are defualt but not SHLTK and ORIEN. SHLTK set to 1 is for considering the shell thickness with 1 being for thickness taken into consideration but rigid bodies are excluded. ORIEN is for reorientation. With ORIEN being set to 2 which enables manual and automatic segments reorientation. Card 2 is all defualt, the same for Card 3. Card 4 is an optional card and is not default with ignore set to 2 it allows initial penetrations to exist rather than trying to eliminate them by moving nodes. FRCENG set to 1 enables calcu-

lation of friction between surfaces and stores it in the INTOR file. SKIPWRG is defualt and is enabled for vizualizing a stationary planar rigid wall. OUTSEG is for posting contact_spotweld to d3hsp file, this is enabled by setting to 1. SPOTSTP, this is for determining if the solver should terminate if spotweld is not present. This is set to 0 for ignoring and deleting the weld. SPOTDEL is set to 1 to delete the spotweld elements. SPOTHIN is an optional thickness scale for preventing premature failure of spotwelds this is a value between 0 and 1. Card 5 is also default. SHLEDG set to 1 assumes shell edges are square and flush with the nodes rather than round. With FTALL set to 1 ensures all forces are output to all matching force transducers rather than just the first one.

```
*CONTROL_CONTACT
$#   slsfac    rwpnal    islchk    shlthk    penopt    thkchg     orien
       0.0       0.0         1         1         1         0         2
$#   usrstr    usrfrc     nsbcs    interm     xpene     ssthk      ecdt   tiedprj
         0         0         0         0       0.0         0         0         0
$#    sfric     dfric       edc       vfc        th     th_sf    pen_sf     ptscl
       0.0       0.0       0.0       0.0       0.0       0.0       0.0       1.0
$#   ignore    frceng   skiprwg    outseg   spotstp   spotdel   spothin
         2         1         0         1         0         1       0.5
$#     isym    nserod    rwgaps
         0         0         1
$#   shledg    pstiff    ithcnt    tdcnof     ftall
         1         0         0         0         1
```

For controlling how energy is calculated and stored in CONTROL_ENERGY. For calculating hourglass energy HGEN is set to 2. RWEN is rigidwall energy and is set to the defualt for calculating energy dispersion and it is stored in GLSTAT. SLNTEN is set to 2 for calculating sliding between parts (sliding interface energy). RYLEN is Rayleigh energy dissipation option with this set to 2 it is also stored. IRGEN is initial geometry energy, with this set to 2 it is calculated and stored. MATEN is the detailed material energies option, with it set to 1 the detalied material energies are not computed, which is the default. DRLEN is drilling energy is set to 1 for not calculating the drilling energy, this is the default. DISEN is the dissipation energy calculation option, this is set to 1 for not calculating this is the default.

```
*CONTROL_ENERGY
$#     hgen      rwen    slnten     rylen     irgen     maten     drlen     disen
         2         2         2         2         2         1         1         1
```

For controlling hourgrlass modes in reduced integration elements. IHQ set to 6 is for Belytschko-Bindeman assumed strain co-rotational stiffness form for solid elements in 2D and 3D. QH is the hourglass coeficient this is set to 0.1.

```
*CONTROL_HOURGLASS
$#      ihq        qh
         6       0.1
```

For determaning the automatic timestep control during implicit analysis. IAUTO, is set to 1 for automatically adjusting the timestep size. ITEOPT is set to 11 (default) which is the amount of optimum equilibrium iteration pr timestep, so 11 equilibriums pr timestep.

ITEWIN is the allowable iteratoin window, this is also the default set to 5. DTMIN is the minimum timestep, set to 0.0. DTMAX is the maximum allowable timestep which is set to curve 4441 ("-" is for steps as a function of time).

```
*CONTROL_IMPLICIT_AUTO
$#   iauto    iteopt    itewin     dtmin     dtmax
        1        11         5       0.0     -4441
```

For determaning which type of analysis the solver should use. With IMFLAG set to 1 the solver only solves implicit(iteration based instead of explicit which calculates the next step from the current step). DT=0.1 is the initial timestep. IMFORM is set to 2 this is the element formulation which retains the original element formulation. NSBS is the defualt set to 1 which is for the number of implicit steps in springback. IGS is the flag that controls initial stress, with this set to 1 it includes initial stresses.

```
*CONTROL_IMPLICIT_GENERAL
$#  imflag       dt0    imform      nsbs       igs
        1       0.1         2         1         1
```

The following keyword is to determine how the implicit solver is calculated. These are all standard and not changed from the standard from the manual.

```
*CONTROL_IMPLICIT_SOLUTION
$#  nsolvr    ilimit    maxref     dctol     ectol     rctol     lstol    abstol
       12        11        15     0.001      0.011.00000E10       0.91.0000E-10
$#   dnorm    diverg     istif   nlprint    nlnorm
        2         1         1         0         2
$#  arcctl    arcdir    arclen    arcmth    arcdmp
        0         0       0.0         1         2
$#   lsmtd     lsdir      irad      srad      awgt      sred    kssize
        4         2       0.0       0.0       0.0       0.0        10
```

This keyword is controlling how the implicit solver solves the linear system of equations at each timestep. LSOLVR set to 2 for using parallel multi-frontal sparse solver for solving the linear equitation system. LPRINT set to 1 for writing how the CPU is being used. NEGEV set to 2 prints warnings when eigenvalues are negative and is the default. ORDER is the default for LS-Dyna. DRCM is for adding drilling stiffness and is the defualt when set to 4. DRCPRM is for how fast the drilling happends and is set to 100 even though no drilling occurs. AUTOSPC set to 1 is for scanning the stiffness matrix for unconstrained, unattached d.o.f, and generates constraints to negate negative eigenvalues. AUTOTOL is the tolerance for singularities if the value is below AUTOTOL a constraint is generated. LCPACK is set to 2 and is the default matrix assembly. MTXDMP is the setting for dumping or not dumping the stiffness matrix, with this being set to 0 no dumping occurs. IPARM1 is the maximum amount of iterative solvers. RPARM1 is for the absolute tolerence for convergence this is set to 0 which is not in use. RPARM2 is for iterative solvers the relative convergence being set to 0.001. RPARM4 is the nominal block size which is set to the default at 240. EMXDMP is for dumping the relative stiffness and mass matrix this set to 0 means no dumping. RDCMEM is for capping the dynamic memomry with it being set to 0.85 it is 85% of available memory since this simulation is running on a workstaion this

is needed for the stability of the system(can be tweaked for more perfoming computers).

```
*CONTROL_IMPLICIT_SOLVER
$#  lsolvr    lprint     negev     order      drcm    drcprm   autospc   autotol
         2         1         2         0         4     100.0         11.10000E-7
$#  lcpack    mtxdmp   iparm1    rparm1    rparm2         -    rparm4
         2         0       500       0.0     0.001                240.0
$#  emxdmp    rdcmem
         0      0.85
```

This keyword is simply for what is being written to the terminal. NPOPT set to 1 supresses nodal coordinates and displacements. NEECHO is for determining what is written in the echo file. NEECHO set to 0 means everything is written. NREFUP is for updating the reference node coordinates, this set to 0 which does not update the reference node.

```
*CONTROL_OUTPUT
$#   npopt    neecho    nrefup
         1         0         0
$#   iprtf    ierode   tet10s8
         0         0         2
```

This keyword is how the computer is supposed to run the simulation. NCPU set to 0 is for the user to determine the amount of threads to be allockated. NUMRHS set to 0 allockaes the same amount of right hand sides as cpus in memory which is the default. CONST is set to 1 which enables consistancy, even though NCPU can be 8 consistancy stays on. PARA set to 0 disables parralel force assembly.

```
*CONTROL_PARALLEL
$#    ncpu    numrhs     const      para
         0         0         1         0
```

For controlling how the solid elements responds in simulaiton. ESORT is for how the sorting of tetrahedral and pentahedral elements to avoid using degenerate formulations with this set to 1, sorting is enabled by sorting tetrahedral elements to type 10 and pentahedral to type 15 and so on. FMATRIX is for determaning the deformation gradient matrix this set to 2 is for computing the the deformation matrix directly. NIPTETS is for how many integration points are needed for quadratic tetrahedron elements with the default being 4. SWLOCL is for how the stresses are represented, with 1 being for them to be in the global coordinate system.

```
*CONTROL_SOLID
$#   esort   fmatrix   niptets    swlocl
         1         2         4         1
```

For controlling what solution is searched for the following keyword is needed. SOLN set to 2 is for solving combined structural, and thermal analysis. NLQ set to 0 uses the default vector length for the solution. ISNAN disables "Not a Number" checking in the force and number arrays. LCINT uses the default of 100 points for material curve rediscreization. LCACC is for truncate curves with LCACC set to 0 this is disabled and no truncating is happening. NCDCF set to 1 evaluates any defined curve functions every cycle.

```
*CONTROL_SOLUTION
$#     soln       nlq     isnan     lcint     lcacc     ncdcf
         2         0         0         0         0         1
```

The purpose for this keyword is to stop the termination. ENDTIM is at what time the simulation stops, so 120 seconds. ENDCYC is at what cycle the termination should stop. This is set to 0 hence it is not in use. DTMIN is the minimum time step. ENDENG is for stopping the simulation at a specific energy change, this is not used. ENDMAS is only viable if mass scaling is enabled, it is not in this simulation therefore it is the default value.

```
*CONTROL_TERMINATION
$#   endtim    endcyc     dtmin    endeng    endmas
     120.0         0       0.0       0.0  100000.0
```

This keyword is for a nonlinear or thermal mechanical simulation. REFMAX is maximum amount of matrix reformations pr step this is set to 100. TOL is the convergence tolerance for temperature, this is the set to 1^{-4}. DCP is divergence control parameter and this is set to 0.5 since it is a transient problem(changing over time). LUMPBC is for Lump thermal boundary conditions, and is disabled. THLSTL enables line search with convergence tolerence of 0.1

```
*CONTROL_THERMAL_NONLINEAR
$#   refmax       tol       dcp    lumpbc    thlstl
       1001.00000E-4       0.5         0       0.1
```

Is for keyword for the thermal solver for either a thermal simulation or a thermal mechanical simulation. ATYPE is the type of thermal analisys this being set to 1 is for transient analisys. PTYPE set to 1 is for a nonlinear problem with the material properties being evaluated at gauss point temperature. SOLVER set to 11 is for direct solver type. GPT is the number of gauss points used in each solid element this is the defualt when set to 0, which results in 8 points. EQHEAT is the eqiivalent of the heat and is set to the default at 1.0. FWORK is fraction of mechanichal work which is converted into heat and is set to the default at 1.0. MSGLVL is for output message type with this set to 0 there is no output into the console. MAXITR is the maximum number of iterations set to the default at 500. ABSTOL is the absolute convergence tolerance and is set to the default. RELTOL is the relative convergence tolerance and is set to the default. OMEGA is relaxation parameter and is set to the default. TSF is thermal speedup factor, this can be used to artificially scale the heating. This is set to the default at 1.0.

```
*CONTROL_THERMAL_SOLVER
$#    atype     ptype    solver         -       gpt    eqheat     fwork
          1         2        11                 0       1.0       1.0
$#   msglvl    maxitr    abstol    reltol     omega    unused    unused       tsf
          0       5001.0000E-101.00000E-6       0.0                           1.0
```

For setting the time step for the heating aspect the following keyword is used. TS is timestep control and is set to 0 which means fixed timestep. TIP is the time integration parameter and is set to 1.0 for fully implicit integration which is the default. ITS is the ini-

tial timestep at 0.1. TMIN is the minimum timestep set to 1.0E-9. TMAX is the maximum timestep set to 0.5. DTEMP is the maximum amount of heat change per step, this is set to 100K. TSCP is the timestep control paremeter which can be between 0 and 1, this is a factor for which the time step is scaled if the simulation should exceed DTEMO. LCTS is the load curve for how the time steps should forego, this is set to a custom curve 4441.

```
*CONTROL_THERMAL_TIMESTEP
$#      ts       tip       its      tmin      tmax     dtemp      tscp      lcts
         0       1.0       0.1    1.0E-9       0.5     100.0      0.95      4441
```

For controlling the structural time step the following keyword is used. For determining the initial time step in LS-DYNA DTINIT is set to 0.1. TSSFAC is a scale factor for load factor this is set to 1. ISDO is for shells and is not useful for solids, this is set to 0. TSLIMIT is for shell element minimum timestep and is not usefull for solids, this is set to 0.0. DT2MS is the timestep size for mass-scaled solutions, this is set to 0 since no mass scale is used. LCTM is a load curve for setting step size set to 4441.

```
*CONTROL_TIMESTEP
$#  dtinit    tssfac      isdo    tslimt     dt2ms      lctm
       0.1       1.0         0       0.0       0.0      4441
```

For keeping the main.k neat a seperate file for boundary conditions is made, the following keyword includes the file.

```
*INCLUDE
boundary.k
*END
```

All the following database inquries for LS-DYNA has the same inputs. These are for the ascii files so these are for bakward compatibilty.

```
*DATABASE_BNDOUT,*DATABASE_DEFORC,*DATABASE_ELOUT,*DATABASE_GLSTAT,*
    DATABASE_JNTFORC,*DATABASE_MATSUM,*DATABASE_SPCFORC,*DATABASE_RCFORC,*
    DATABASE_NODOUT,*DATABASE_NCFORC, *DATABASE_TPRINT
$#      dt    binary      lcur     ioopt
      0.03         3         0         1
```

This is the timestep for the d3plot file. For determining how "smooth" the animation looks the timestep for the animation is DT set to 0.5.

```
*DATABASE_BINARY_D3PLOT
$#      dt
       0.5
```

This is for adding material, part, eqation of stae, section and hourglass data to the d3plot file. IFILE set to 2 outputs data to d3prop. IMATL set to 1 outputs data hoirglass, part and section data to the plot file. IWALL is for outputting *RIGIDWALL data which is set to 0.

```
*DATABASE_BINARY_D3PROP
$#   ifile     imatl     iwall
         2         1         0
```

For controlling the plotting databases the following keyword is used. NEIPH, NEIPS and

MAXINT arent affected for plotting. STRFLAG set to 10 is enables plastic strain tensor to output to the d3plot but not strain or thermal strain tensors. SIGFLIG is for including the stress tensor for shells this is set to 1 as default and includes them. EPSFLG flag for including effective plastic strains is default and includes them. RLTFLG for including stress resultants for shells is set to 1 for inclusion. ENGLFG is flag for internal energy density in shell thickness and is included even though there are no shells. CMPFLG is for indicate coordinate system as output from stress and strain this is set to 0 for setting this to the global coordinate system. IEVERP is for determening multiple d3plot files or a single d3plot file this is set to 0 for multiple. BEAMIP is number of beam integretions for output, no beams are in this simulation so this is kept at 0 as default. DCOMP is data compression for eliminating rigid body data this is turned of as indicated by it being set to 1. SHGE set to 1 is for not including hourglass energy density in shells. STSSZ is for including shelle element time steps or added mass this is turned off with it being set to 1. N3THDT is for including material energy in the d3thdt database file, this is set to 2 for the output written to the d3thdt file. NINTSLD uses the default integration points written to ls dyna. PKP_SEN is a flag to output peak pressure and surface energy, with this set to 0 it is turned off. SCLP is a smoothing factor for the preasure which is turned off by it being 0.0. HYDRO set to 0 means no additinal shock physics history. MSSCL is for data mass scaling, with this being set to 2 it is outputtig a percantage increase in nodal mass (there is no mass scaling so this is not used). THERM is the outputiing of thermal data, this is set to 2 for outputting temperature and flux.

```
*DATABASE_EXTENT_BINARY
$#   neiph     neips    maxint    strflg    sigflg    epsflg    rltflg    engflg
         0         0         0        10         1         1         1         1
$#  cmpflg    ieverp    beamip     dcomp      shge     stssz    n3thdt
         0         0         0         1         1         1         2
$# nintsld   pkp_sen      sclp     hydro     msscl     therm
         0         0       0.0         0         2         2
```

The following keyword is for defining the output of binary files. This is set to the defualt. IFORM set to 0 means that the database format is LS-DYNA standard database format. IBINARY set to 0 is for 64 bit format for binary outputs and is the default when using double precission(as the simulation is).

```
*DATABASE_FORMAT
$#   iform   ibinary
         0         0
```